

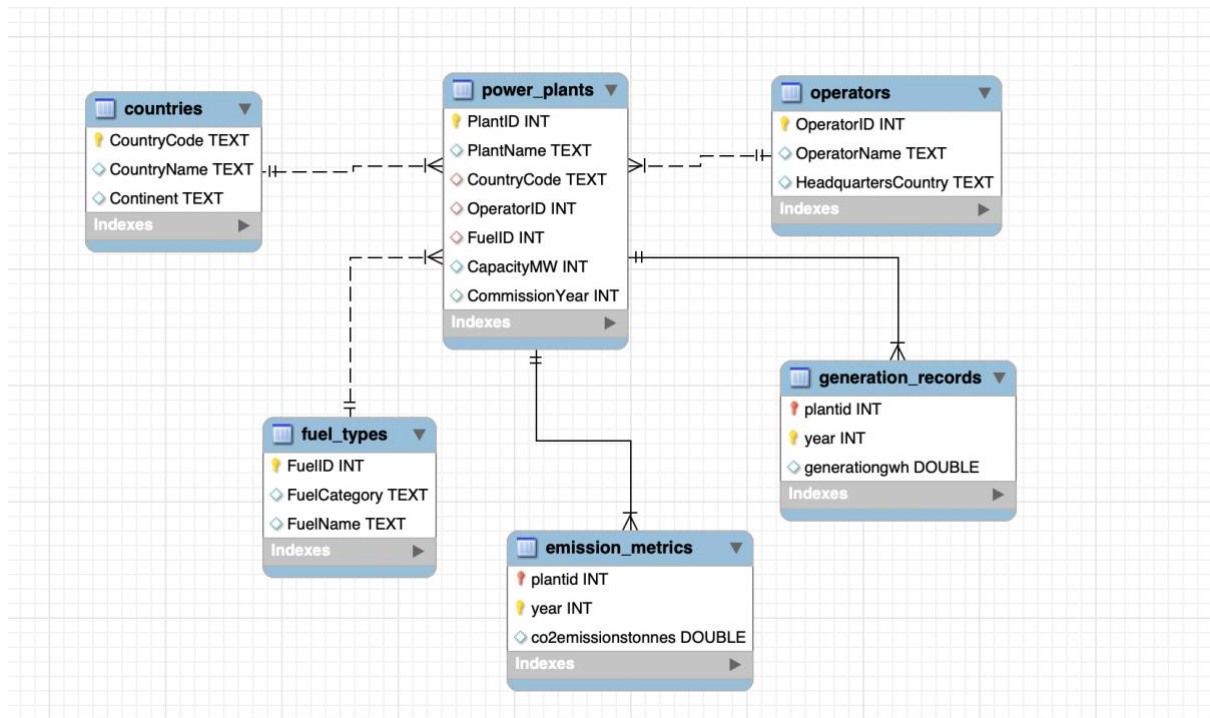
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Homework #3

EER Diagram



SQL Problems

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13 -- 1) Retrieve the plant name, country name, operator name, fuel category, fuel name,  
14 -- capacity (in MW), and commission year for all power plants without using table aliases.  
15 -- Sort the results in descending order by capacity.
```

```
16  
17 • select  
18     power_plants.PlantName,  
19     countries.CountryName,  
20     operators.OperatorName,  
21     fuel_types.FuelCategory,  
22     fuel_types.FuelName,  
23     power_plants.CapacityMW,  
24     power_plants.CommissionYear  
25 from power_plants natural join countries  
26         natural join operators  
27         natural join fuel_types  
28 order by power_plants.CapacityMW desc;  
29
```

100% 39:28

Result Grid Filter Rows: Search Export:

PlantName	CountryName	OperatorName	FuelCategory	FuelName	CapacityMW	CommissionYear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

```
30 -- 2) Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all  
31 -- power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.
```

```
32  
33 • select  
34     power_plants.PlantName,  
35     power_plants.CountryCode,  
36     generation_records.year,  
37     generation_records.generationgwh  
38 from power_plants natural join generation_records  
39 where generation_records.year = 2024  
40 order by generation_records.generationgwh desc;  
41
```

100% 48:40

Result Grid Filter Rows: Search Export:

PlantName	CountryCode	year	generationgwh
Three Gorges	CHN	2024	98800
Itaipu	BRA	2024	89000.1
Tuoketuo	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariwa	JPN	2024	0

Result Grid
Form Editor

```

42 -- 3) Retrieve the plant name, country code, calendar year, annual generation (in GWh),
43 -- and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases.
44 -- Sort the results in ascending order by CO2 emissions.
45

```

```

46 • select
47     power_plants.PlantName,
48     power_plants.CountryCode,
49     generation_records.year,
50     generation_records.generationgwh,
51     emission_metrics.co2emissionstonnes
52 from power_plants natural join generation_records
53     natural join emission_metrics
54 where generation_records.year = 2024
55 order by emission_metrics.co2emissionstonnes asc;
56

```

100% 57:44

Result Grid Filter Rows: Search Export:

PlantName	CountryCode	year	generationgwh	co2emissionstonnes
Grand Coulee	USA	2024	21500.5	0
Itaipu	BRA	2024	89000.1	0
Kashiwazaki-Kariwa	JPN	2024	0	0
Three Gorges	CHN	2024	98800	0
Loy Yang	AUS	2024	14200.4	1950000
Janschwalde	DEU	2024	18400.2	2450000
Vindhyachal	IND	2024	35600.8	3100000
Tuoketuo	CHN	2024	71000.3	5200000

Result Grid Form Editor

```

57 -- 4) Write a SQL query using a Common Table Expression (CTE) to calculate the total cumulative
58 -- power generation (in GWh) for each operator across all available years without using table aliases.
59 -- Retrieve the operator name, headquarters country, and their total generated power, and sort the results
60 -- in descending order by total generation.
61

```

```

62 • with cumulative_power_gen as(
63     select
64         power_plants.OperatorID,
65         sum(generationgwh) as cumul_power_gen
66     from power_plants natural join generation_records
67     group by power_plants.OperatorID
68 )
69
70 select
71     operators.OperatorName,
72     operators.HeadquartersCountry,
73     cumulative_power_gen.cumul_power_gen
74 from operators join cumulative_power_gen on operators.OperatorID = cumulative_power_gen.OperatorID
75 order by cumulative_power_gen.cumul_power_gen desc;
76

```

100% 52:75

Result Grid Filter Rows: Search Export:

OperatorName	HeadquartersCountry	cumul_power_g...
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

Result Grid Form Editor

```

79  -- 5) Write a SQL query using two separate Common Table Expressions (CTEs)
80  -- one to calculate total power generation by country code and another to calculate total
81  -- CO2 emissions by country code without using table aliases. Then, join these CTEs with
82  -- the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes).
83  -- Sort the results in descending order by total generation.
84  with country_total_power_gen as(
85      select
86          sum(generation_records.generationgwh) as total_generationgwh,
87          power_plants.CountryCode
88      from generation_records natural join power_plants
89      group by power_plants.CountryCode
90  ),
91
92  country_total_emissions as(
93      select
94          sum(emission_metrics.co2emissionstonnes) as total_emissions,
95          power_plants.CountryCode
96      from emission_metrics natural join power_plants
97      group by power_plants.CountryCode
98  )
99
100  select
101      countries.CountryName,
102      country_total_power_gen.total_generationgwh,
103      country_total_emissions.total_emissions
104  from countries natural join country_total_power_gen
105       natural join country_total_emissions
106  order by country_total_power_gen.total_generationgwh desc;
107

```

50% 59:106

Result Grid  Filter Rows: Export: 

CountryName	total_generationgwh	total_emissions	
China	169800.3	5200000	
Brazil	89000.1	0	
India	35600.8	3100000	
United States	21500.5	0	
Germany	18400.2	2450000	
Australia	14200.4	1950000	
Japan	0	0	